

# A disproof of Conjecture 00000007869

The random-order lower bound for all deterministic algorithms contradicts the stated upper bound for Best Fit

## 1 The conjecture

Conjecture 00000007869 reads (English):

*Definition:* The random-order competitive ratio of online bin packing: the expected competitive ratio  $R_{ro}^n$  under a uniformly random arrival order. *Conjecture:* Every deterministic online algorithm has  $R_{ro}^n \geq 3/2 + \epsilon_0$  for an explicit positive  $\epsilon_0$ , while Simple-Best-Fit achieves  $R_{ro}^n \leq 3/2 + O(n^{-1/2})$ ; and the optimal separation of the random-order advantage over the worst order is the explicit jump from  $4/3$  to  $3/2$ , determined by extremal instances of bimodal volume distributions.

The Chinese text agrees. It states the first clause for all deterministic online algorithms and calls  $\epsilon_0$  an explicit positive constant.

We show that the first two clauses contradict each other. Best Fit is a deterministic online algorithm, so clause 1 applies to it. The final clause (the “jump from  $4/3$  to  $3/2$ ”) is not formalized and nothing is claimed about it.

## 2 Readings

Write  $R(A, n)$  for  $R_{ro}^n(A)$  and  $A_0$  for Simple-Best-Fit.

- **Clause 1.** There is a constant  $\epsilon_0 > 0$  with  $R(A, n) \geq 3/2 + \epsilon_0$  for every deterministic online algorithm  $A$ . The constant does not depend on  $n$  (it is called a constant). We allow it to depend on  $A$ ; this only weakens the clause.
- **Clause 2.** There is a constant  $C \in \mathbb{R}$  with  $R(A_0, n) \leq 3/2 + C/\sqrt{n}$  for all sufficiently large  $n$ . This is the standard meaning of  $3/2 + O(n^{-1/2})$ , and it is implied by the “for all  $n \geq 1$ ” reading.

The quantifier on  $n$  in clause 1 is not explicit, so we refute each of these readings: for all  $n \geq 1$ ; for all sufficiently large  $n$ ; for infinitely many  $n$ ; as a limit ( $R(A, n) \rightarrow L \geq 3/2 + \epsilon_0$ ); as a limit superior ( $\limsup_n R(A, n) \geq 3/2 + \epsilon_0$ ). We also refute clause 1 for all  $n \geq 1$ , for all sufficiently large  $n$ , or as a limit, against the weaker clause 2 “for infinitely many  $n$ ”.

**Not refuted.** (a) Clause 1 read as “for some  $n$ ”. (b) Clause 1 and clause 2 both read only “for infinitely many  $n$ ” (the two sets of  $n$  could be disjoint). (c) An  $\epsilon_0$  that depends on  $n$ ; the text excludes this by calling  $\epsilon_0$  a constant.

**Convention.** In Lean  $x/0 = 0$ . This matters only for  $n = 0$  (empty instances), which no reading uses.

### 3 The clash for an arbitrary ratio

**Lemma 1** (`eventually_div_sqrt_lt`). *For every  $C \in \mathbb{R}$  and  $\epsilon > 0$ ,  $C/\sqrt{n} < \epsilon$  for all sufficiently large  $n$ .*

*Proof.*  $\sqrt{n} \rightarrow \infty$ , so  $C/\sqrt{n} \rightarrow 0$ . Explicitly,  $n > (\max(C, 0)/\epsilon)^2$  suffices.  $\square$

**Theorem 2.** *Let  $\mathcal{A}$  be any set,  $R : \mathcal{A} \times \mathbb{N} \rightarrow \mathbb{R}$  any function and  $A_0 \in \mathcal{A}$ ,  $\epsilon_0 > 0$ ,  $C \in \mathbb{R}$ .*

- (i) *If  $R(A_0, n) \geq 3/2 + \epsilon_0$  for infinitely many  $n$  and  $R(A_0, n) \leq 3/2 + C/\sqrt{n}$  for all large  $n$ , we get a contradiction (`clash_frequently_eventually`).*
- (ii) *If  $R(A_0, n) \geq 3/2 + \epsilon_0$  for all large  $n$  and  $R(A_0, n) \leq 3/2 + C/\sqrt{n}$  for infinitely many  $n$ , we get a contradiction (`clash_eventually_frequently`).*
- (iii) *If  $R(A_0, n) \rightarrow L \geq 3/2 + \epsilon_0$  and  $R(A_0, n) \leq 3/2 + C/\sqrt{n}$  for infinitely many  $n$ , we get a contradiction (`clash_tendsto_frequently`).*
- (iv) *If  $R(A_0, \cdot)$  is bounded below,  $\limsup_n R(A_0, n) \geq 3/2 + \epsilon_0$  and  $R(A_0, n) \leq 3/2 + C/\sqrt{n}$  for all large  $n$ , we get a contradiction (`clash_limsup_eventually`).*

*Proof.* (i), (ii): by the lemma there is an  $n$  with  $3/2 + \epsilon_0 \leq R(A_0, n) \leq 3/2 + C/\sqrt{n} < 3/2 + \epsilon_0$ . (iii):  $R(A_0, n) \geq 3/2 + \epsilon_0/2$  for all large  $n$ ; apply (ii) with  $\epsilon_0/2$ . (iv): eventually  $R(A_0, n) \leq 3/2 + C/\sqrt{n} \leq 3/2 + \epsilon_0/2$ , so  $\limsup_n R(A_0, n) \leq 3/2 + \epsilon_0/2 < 3/2 + \epsilon_0$ . Lower boundedness is what Mathlib's `limsup_le_of_le` needs for real sequences.  $\square$

**Corollary 3** (`clauses_inconsistent`). *For any  $R$ , any class  $\text{IsDetOnline} \subseteq \mathcal{A}$  and any  $A_0$  in the class, the following two statements are not both true:*

- *for every  $A$  in the class there is  $\epsilon_0 > 0$  with  $R(A, n) \geq 3/2 + \epsilon_0$  for infinitely many  $n$ ;*
- *there is  $C$  with  $R(A_0, n) \leq 3/2 + C/\sqrt{n}$  for all large  $n$ .*

So neither the exact definition of  $R_{ro}^n$  nor the exact meaning of the nonstandard name “Simple-Best-Fit” matters, provided both clauses use the same  $R_{ro}^n$  and the named algorithm is deterministic and online.

### 4 Concrete online bin packing

The Lean file also gives concrete definitions and states the main theorem for them.

**Definition 4.** An  $n$ -item instance  $I$  is a size function  $s : \{0, \dots, n-1\} \rightarrow (0, 1]$ ; bins have capacity 1. A deterministic online algorithm (`OnlineAlg`) is a function that, when an item of size  $s$  arrives, sees the sizes of the earlier items (in arrival order), the current bin loads and  $s$ , and returns a bin index. An out-of-range index, or a bin where the item does not fit, means “open a new bin”. The decision never depends on future items. `binsUsed`( $A, \cdot$ ) is the number of bins opened on an arrival sequence. **Best Fit** (`bestFit`) puts the item in the fullest open bin in which it fits, taking the lowest index among ties, and opens a new bin if it fits nowhere. `OPT`( $I$ ) (`opt`) is the least  $k$  such that some map  $f : \{0, \dots, n-1\} \rightarrow \{0, \dots, k-1\}$  has  $\sum_{f(i)=b} s(i) \leq 1$  for every  $b$ . For a permutation  $\sigma$ , the arrival sequence is  $s(\sigma(0)), \dots, s(\sigma(n-1))$ , and

$$\text{expRatio}(A, I) = \frac{\frac{1}{n!} \sum_{\sigma} \text{binsUsed}(A, I_{\sigma})}{\text{OPT}(I)}, \quad R_{ro}^n(A) = \sup_{I \text{ with } n \text{ items}} \text{expRatio}(A, I).$$

The proofs use only that  $R_{ro}^n \geq 0$  (`Rro_nonneg`, needed for the limsup reading).

**Theorem 5** (`conjecture_7869_false`).  $\neg(\text{Clause1} \wedge \text{Clause2})$ , where *Clause1*: for every *OnlineAlg*  $A$  there is  $\epsilon_0 > 0$  with  $R_{ro}^n(A) \geq 3/2 + \epsilon_0$  for infinitely many  $n$ ; and *Clause2*: there is  $C$  with  $R_{ro}^n(\text{bestFit}) \leq 3/2 + C/\sqrt{n}$  for all large  $n$ .

Since “for all  $n \geq 1$ ” implies “for all large  $n$ ”, which implies “for infinitely many  $n$ ”, this covers the stronger readings. The Lean file also states them separately: `conjecture_7869_false_forall` (one  $\epsilon_0$  for all  $A$  and all  $n \geq 1$ ; one  $C$  for all  $n \geq 1$ ), `conjecture_7869_false_eventually_frequently`, `conjecture_7869_false_limsup` and `conjecture_7869_false_tendsto`.

## 5 Lean formalization

File `lean/Conjecture7869/Basic.lean` (192 lines, Lean 4.33.1, Mathlib v4.33.1), namespace `C7869`. Main theorem: `C7869.conjecture_7869_false : ~(Clause1 /\ Clause2)`. `#print` axioms for each of `conjecture_7869_false`, `conjecture_7869_false_forall`, `conjecture_7869_false_eventually_frequently`, `conjecture_7869_false_limsup`, `conjecture_7869_false_tendsto` and `clauses_inconsistent` reports only `propext`, `Classical.choice` and `Quot.sound`. There is no `sorry` and no `native_decide`.